

Lab Assignment: Bank Account Management System

Submission Details

- **Submission Platform:** Google Classroom
- **Submit:**
 - Your Java file (`AccountTest.java`)
 - At least 2 screenshots showing program execution

Objective

The goal of this assignment is to develop a simple **Bank Account Management System** using Java. Students will practice the following concepts:

- Classes and Objects
- Constructors
- Instance Variables
- Methods
- Decision Structures (if/else)
- Loop Structures
- User Input using Scanner

Program Description

In this assignment, you will create a program that simulates basic bank account operations. The system should allow a user to:

- Deposit money
- Withdraw money
- Display the current balance
- Exit the program

Program Requirements

Account Class

Create a class named **Account** that represents a bank account.

The class must contain the following instance variables:

- **name** : String representing the account holder's name
- **balance** : double representing the current balance

The class must include a constructor that initializes these variables.

- If the initial balance entered by the user is negative, the balance should be set to **0**.

Required Methods

Your **Account** class must implement the following methods:

- **deposit(double amount)** Adds money to the account balance. If the amount is less than or equal to zero, display the message:

```
1 Invalid deposit amount!
```

- **withdraw(double amount)** Subtracts money from the account balance.
 - Withdrawal is allowed only if the amount is positive.
 - The withdrawal amount must not exceed the current balance.
 - If the balance is insufficient, display:

```
1 Insufficient balance!
```

- **getBalance()** Returns the current account balance.
- **getName()** Returns the account holder's name.

AccountTest Class

Create another class named **AccountTest** that contains the **main** method.

- Use the **Scanner** class to obtain user input.
- Ask the user to enter:
 - Account holder name
 - Initial balance
- Create an **Account** object using this information.

Menu System

After creating the account, display the following menu repeatedly:

```
1 1 - Deposit Money
2 2 - Withdraw Money
3 3 - Show Balance
4 4 - Exit
```

The program must behave as follows:

- If the user selects **1** Ask for the deposit amount and call the **deposit()** method.
- If the user selects **2** Ask for the withdrawal amount and call the **withdraw()** method.
- If the user selects **3** Display the account holder name and current balance.
- If the user selects **4** Terminate the program.

The program must continue running using a **loop structure** until the user selects the exit option.

Example Menu Output

```
1 Enter account holder name: Alice
2 Enter initial balance: 1000
3
4 1 - Deposit Money
5 2 - Withdraw Money
6 3 - Show Balance
7 4 - Exit
8
9 Select option: 1
10 Enter deposit amount: 200
```

Technical Requirements

- Use **at least one class and one main class**.
- Implement the required methods in the **Account** class.
- Use **if/else decision structures**.
- Use a **loop structure** to keep the menu running.
- Use the **Scanner** class for user input.